IP addressing

Tasks :-

1.Find the IP address of your computer or mobile device and write down whether it is IPv4 or IPv6. Also mention if it is a public or private IP.

Ans :-

My computer's IPv4 address is **10.79.173.31**. It is an IPv4 private IP address because it belongs to the private **10.0.0.0/8** range. My computer also has an IPv6 address: **2409:40c1:411b:2650:bd39:4c56:4e93:9356.**

**2.Given the following list of IP addresses, identify which ones are valid private IPs and which are public: 192.168.1.10, 10.0.0.5, 172.16.5.20, 8.8.8.8, 123.45.67.89.

Hint: Check the standard private IP address ranges for IPv4.**

Ans :-

➢ IPv4 Private vs Public IP Addresses

The standard **private IPv4 ranges** are:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

IP Address	Type	Reason
`192.168.1.10`	Private	Within `192.168.0.0/16`
`10.0.0.5`	Private	Within `10.0.0.0/8`
`172.16.5.20`	**Private**	Within `172.16.0.0/12`
`8.8.8.8`	**Public**	Not in any private range
`123.45.67.89`	**Public**	Not in any private range

**3.Write a short script in Python or JavaScript that takes an IP address as input and tells the user whether it is a valid IPv4 address or not.

Constraint: Do not use any third-party libraries for validation.**

Ans :-

```
def is_valid_ipv4(ip):
    parts = ip.split(".")
    # IPv4 must have exactly 4 parts
    if len(parts) != 4:
        return False
    for part in parts:
        # Each part must contain only digits
        if not part.isdigit():
            return False
        # Convert to number
        number = int(part)
        # Each number must be between 0 and 255
        if number < 0 or number > 255:
            return False
    return True
# Take IP address from user
ip = input("Enter an IPv4 address: ")

if is_valid_ipv4(ip):
    print("Valid IPv4 address")
else:
    print("Invalid IPv4 address")
```

4.Explain with an example how Instagram or WhatsApp might use IP addresses to detect suspicious login activity from different locations.

Ans :-

- **Suspicious Login Detection Using IP Addresses Understand this images thoroughly:-**

